READING PASSAGE 2

You should spend about 20 minutes on Questions 14–26, which are based on Reading Passage 2 below.

The internal body clock

Each of our cells has an internal 'clock' which dictates our daily rhythms. But why?

A From buffalo to bacteria, oaks to algae, all life follows the same relentless 24-hour cycle, which is driven by the rising and setting of the Sun. Or is it? Over 250 years ago, a French scientist performed a simple experiment that blew apart the idea that all life on Earth does the bidding of the Sun. Intrigued by the way some flowers open and close their leaves each day, Jean-Jacques de Mairan put a heliotrope in a darkened room and observed the effect. He was expecting the plant, robbed of sunlight, to cease its daily routine. To his astonishment, its leaves continued to open and close as if in response to some invisible timekeeper.

B But what – and where – is this internal timekeeper? And how does it achieve such astonishing regularity? These are the mysteries at the forefront of chronobiology, the study of the effect of time in living organisms. The search for answers is about more than just tying up some scientific loose ends. Internal ‘clocks’ clearly cause a lot of hardship. When an individual’s ‘clock’ is knocked out of synchronization through having to work at night or go on long-haul flights, that person can be left utterly unable to think or act. Yet even when these ‘clocks’ work correctly, the natural rhythms of alertness they generate prove dangerous: traffic accident statistics show two deadly peaks, at 4 a.m. and again twelve hours later when people are at their least alert.

C During the 1970s, best-selling books began to emerge claiming such phenomena were manifestations of so-called biorhythms, a set of three cycles governing physical, emotional and intellectual traits. Said to start from the moment of birth and repeat every 23, 28 and 33 days respectively, they were supposed to result in ‘critical days’ when one or more cycles led to sub-optimal performance, with unfortunate consequences.

D Scientists have since dismissed biorhythms as pseudoscience, insisting the existence of the three cycles has no basis in fact. In 1998, Dr Terence Hines of Pace University, New York State published the most comprehensive study of the claims made for biorhythms, reviewing the results of over 130 investigations. He found that three-quarters of them failed to provide any support for biorhythms. Most of the remainder contained blunders ranging from faulty mathematics to basic errors in statistics, while the handful of positive studies were explicable as flukes.

E But there is no doubting the existence of 24-hour biological cycles – or, rather, ones roughly 24 hours long. Following de Mairan's pioneering work, other researchers found that when deprived of the cues provided by sunlight, organisms settle down to 'free-running' cycles that are close to, but rarely exactly, 24 hours long. In the case of humans, the cycle is typically around 24.5 hours long. This is the so-called 'circadian' cycle (from the Latin meaning 'about a day'), the length of which is generated by the mysterious internal 'clock'.

F During the late 1960s, chronobiologists believed they had found the 'clock', in the form of the suprachiasmatic nucleus (SCN), a collection of nerves in a region of the brain known as the hypothalamus. Linked to photosensitive cells in the eye, the SCN senses daylight and triggers the release of hormones like melatonin, which keep body functions in synchronization with the time of day. For a few years the SCN was regarded as the ultimate pacemaker – at least in higher organisms such as ourselves. But in 1971, scientists at the California Institute of Technology working with fruit flies found evidence for something truly amazing. Fruit flies appeared to have genes affecting the daily rhythm of their behavior – suggesting that there are 'clocks' inside each of their cells. Further evidence for this emerged in 1995, when researchers at Massachusetts General Hospital isolated nerve cells from the SCN, and found they kept up a circadian cycle without help from daylight. Finally in 1997, scientists at Northwestern University, Illinois, found a gene that regulates the daily rhythms of cellular activity in mammals.

G The details of how this gene, known as CLOCK, actually works are still being investigated, but they could culminate in better ways of coping with shift-work and jet-lag. Professor Takahashi, who led the research, says that now that we know that circadian clocks exist throughout our bodies, we will need new strategies and therapeutics to reset all of our cells. But how do all the biochemical 'clocks' inside their cells stay in synchronization and why did organisms bother to acquire a link with sunlight? So far, no-one knows, though there are several theories. For example, sunlight might be useful in keeping the myriad cellular 'clocks' in lockstep.

H In the search for answers, researchers are studying the behavior of a microbe called cyanobacterium. In 2005, a team at Nagoya University, Japan, showed that chemical reactions between three proteins produced by this bacterium ebbed and flowed on a 24-hour cycle. Cyanobacterium is the oldest form of life on Earth, so this suggests that cellular 'clocks' have existed for over three billion years. All subsequent organisms have followed suit, says Professor Johnson of Vanderbilt University, Tennessee. 'Bacteria, fungi, plants and animals all appear to have evolved clock systems independently from each other.' Quite why is still unknown. But one thing is clear: the daily routine of life is certainly not a modern invention.

Questions 14–18

Reading Passage 2 has eight paragraphs, ***A–H****.*

Which paragraph contains the following information?

Write the correct letter, ***A–H****, in boxes 14–18 on your answer sheet.*

14 support for an earlier finding from experiments on insects

15 criticism of the way some research data was analysed

16 reference to a basic test on a plant

17 a statement that the length of natural cycles varies slightly between living things

18 reference to a theory that became popular with the public

Questions 19–22

Complete the summary below.

Choose ***NO MORE THAN TWO WORDS*** *from the passage for each answer.*

Write your answers in boxes 19–22 on your answer sheet.

An overview of research into the existence of the internal ‘clock’

Towards the end of the 1960s, research into the SCN, which is located in the hypothalamus, led chronobiologists to believe they had found the internal ‘clock’ at last. Their theory was based on the fact that certain hormones that regulate the timing of physical functions are controlled by the SCN. The SCN is able to perceive **19** _____ as a result of connections to the eye.

Several years later, research involving **20** _____ showed that the timing of their normal actions was regulated by certain **21** _____, or ‘clocks’, in their cells. Thus the internal ‘clock’ is not an individual mechanism, as previously thought, but exists within each cell. Other scientists went on to conduct experiments in the dark that proved that a **22** _____ was maintained by individual SCN cells.

Questions 23 and 24

*Choose **TWO** letters, **A–E**.*

Write the correct letters in boxes 23 and 24 on your answer sheet.

Which **TWO** of the following problems are stated as being linked with the internal ‘clock’?

A a decrease in output in workplaces without windows to let in natural light
B a failure among workers on night-shifts to perform tasks
C an increase in casualties on the roads at specific times of day
D a decline in efficiency on certain days in a number of fixed cycles
E a rise in mood swings dependent on the appearance of the Sun

Questions 25 and 26

*Choose **TWO** letters, **A–E**.*

Write the correct letters in boxes 25 and 26 on your answer sheet.

Which **TWO** of the following issues relating to internal ‘clocks’ remain to be solved, according to information in the passage?

A the precise length of the free-running daily cycle in different animals
B the reason for ‘clocks’ being temporarily driven out of synchronisation
C the reason for living beings forming a connection with the Sun’s rhythms
D the way in which all ‘clocks’ in a single organism are able to stay in step with one another
E the location of the ‘clocks’ across the whole range of living organisms

Disclaimer
Compiled, formatted, and lightly proofread by ZYZ Reading Walks.
All copyright in the underlying works remains with the original authors and publishers.
No affiliation with or endorsement by any rights holder (including IELTS® owners).
For non-commercial educational use only. This notice must remain intact in all copies.
Available free of charge from ZYZ Reading Walks. Resale or any paid distribution is prohibited.

Questions 14–18 段落信息匹配题

题号	答案	题干翻译	详细定位与翻译	详细解释
14	**F**	为早先通过昆虫实验所得出的发现提供支持性证据	**第F段：** *Fruit flies appeared to have genes affecting the daily rhythm of their behavior — suggesting that there are 'clocks' inside each of their cells. Further evidence for this emerged in 1995, when researchers at Massachusetts General Hospital isolated nerve cells from the SCN, and found they kept up a circadian cycle without help from daylight.* 果蝇似乎拥有一些能够影响其行为日常节律的基因，这表明果蝇的每个细胞内部都存在“生物钟”。1995年，人们发现了支持这一观点的进一步证据：马萨诸塞州总医院的研究人员从视交叉上核中分离出神经细胞，发现这些细胞即使没有日光帮助，也能继续维持昼夜节律。	**earlier finding from experiments on insects** 对应1971年对 **fruit flies 果蝇**的研究发现；**support for** 对应后面的 **Further evidence for this emerged**。 这里尤其要注意，题目问的不是“昆虫实验本身”，而是“对昆虫实验早期发现的支持”。1995年的细胞实验进一步证明：生物钟并不只存在于SCN这个整体结构中，而可能存在于单个细胞内部。 同义替换： · support＝**further evidence** · earlier finding＝1971年的发现 · insects＝**fruit flies**
15	**D**	对某些研究数据分析方式的批评	**第D段：** *Most of the remainder contained blunders ranging from faulty mathematics to basic errors in statistics, while the handful of positive studies were explicable as flukes.* 其余大多数研究都存在错误，从数学计算有误到基本的统计错误不等；而少数得出正面结果的研究也可以被解释为偶然现象。	题目中的 **the way some research data was analysed**，指研究者如何用数学和统计方法处理数据。 原文批评这些研究中存在： · **faulty mathematics**：错误的数学计算； · **basic errors in statistics**：基本的统计错误。 这正是对“数据分析方式”的批评。**blunders** 表明作者的评价非常负面。
16	**A**	提到一次针对植物进行的基础、简单实验	**第A段：** *Over 250 years ago, a French scientist performed a simple experiment … Jean-Jacques de Mairan put a heliotrope in a darkened room and observed the effect.* 250多年前，一位法国科学家进行了一项简单实验……让-雅克·德梅朗把一株向日性植物放进一间黑暗的房间，并观察实验效果。 *To his astonishment, its leaves continued to open and close as if in response to some invisible timekeeper.* 令他惊讶的是，这株植物的叶片仍然继续张开和闭合，仿佛在回应某种看不见的计时器。	题目中的 **a basic test** 对应原文的 **a simple experiment**；**on a plant** 对应德梅朗将一株 **heliotrope** 放入黑暗房间。 实验的基本逻辑是：去掉阳光这一外部提示，观察植物是否仍保持原有节律。结果植物依然每天开合叶片，由此暗示生命体内部可能存在独立的计时机制。
17	**E**	说明不同生物的自然周期长度存在轻微差异	**第E段：** *When deprived of the cues provided by sunlight, organisms settle down to 'free-running' cycles that are close to, but rarely exactly, 24 hours long. In the case of humans, the cycle is typically around 24.5 hours long.* 当失去阳光所提供的提示时，生物会进入“自由运行”的周期。这些周期接近24小时，但很少恰好是24小时。对于人类而言，这一周期通常约为24.5小时。	**varies slightly** 对应 **close to, but rarely exactly, 24 hours**：都接近24小时，但并不完全相同。 **between living things** 对应原文中的一般生物 **organisms**，以及随后举出的具体例子 **humans：24.5 hours**。这说明不同生命体的自由运行周期会有轻微差别。 不能选C段的23、28、33天，因为那些属于后来被否定的“生物节律理论”，而本题说的是实际存在的自然周期。
18	**C**	提到一种曾经在公众中流行的理论	**第C段：** *During the 1970s, best-selling books began to emerge claiming such phenomena were manifestations of so-called biorhythms, a set of three cycles governing physical, emotional and intellectual traits.* 20世纪70年代，一些畅销书开始出现，声称这些现象是所谓“生物节律”的表现。这种理论认为有三个周期分别控制人的身体、情绪和智力特征。	题目中的 **a theory** 指 **biorhythms 生物节律理论**；**became popular with the public** 没有直接写成“受公众欢迎”，而是通过 **best-selling books 畅销书**体现。 书能成为畅销书，就说明这一理论曾在普通公众中广泛传播并受到关注。 注意D段讲的是科学家后来如何否定该理论；“曾经流行”出现在C段。

Questions 19–22 摘要填空题

题目摘要翻译

20世纪60年代末，对位于下丘脑的SCN进行的研究，使时间生物学家认为他们终于找到了人体内部的“生物钟”。他们的理论建立在这样一个事实之上：某些调节身体功能时间安排的激素由SCN控制。由于SCN与眼睛相连，它能够感知第19空。几年后，一项以第20空为对象的研究发现，它们正常行为的时间安排受到细胞中某些第21空，也就是“生物钟”的调节。由此可见，内部生物钟并不是先前认为的一个单独机制，而是存在于每一个细胞中。之后，其他科学家在没有光照的情况下进行实验，证明单独的SCN细胞仍然能够维持一个第22空。

题号	答案	题干翻译	详细定位与翻译	详细解释
19	**daylight**	由于与眼睛相连，SCN能够感知_____。	**第F段：** *Linked to photosensitive cells in the eye, the SCN senses daylight and triggers the release of hormones like melatonin.* SCN与眼睛中的感光细胞相连，它能够感知日光，并促使褪黑素等激素释放。	题干中的： • **connections to the eye**＝*linked to photosensitive cells in the eye*； • **perceive**＝*senses*。 因此被感知的对象就是 **daylight 日光**。 虽然文章其他地方也出现了 **sunlight**，但本题定位句直接是 **senses daylight**，所以最稳妥、最精确的答案是 **daylight**。
20	**fruit flies**	几年后，以_____为研究对象的研究表明……	**第F段：** *But in 1971, scientists at the California Institute of Technology working with fruit flies found evidence for something truly amazing.* 但是在1971年，加州理工学院研究果蝇的科学家发现了某种非常惊人的证据。	摘要前面讲的是20世纪60年代末；**Several years later** 对应后面的 **in 1971**。 题干 **research involving…** 对应原文 **scientists … working with…**，研究对象是 **fruit flies 果蝇**。 必须保留复数形式，答案为两个词：**fruit flies**。
21	**genes**	它们正常行为的时间安排受到细胞中某些_____，也就是“生物钟”的调节。	**第F段：** *Fruit flies appeared to have genes affecting the daily rhythm of their behavior – suggesting that there are ‘clocks’ inside each of their cells.* 果蝇似乎拥有一些影响其行为日常节律的基因，这表明它们的每个细胞内部都存在“生物钟”。	题干中的： • **the timing of their normal actions**＝*the daily rhythm of their behavior*； • **was regulated by**＝*genes affecting*。 因此答案是 **genes**。 从语法上看，前面有 **certain**，后面可以接复数可数名词；原文中也使用了复数 **genes**。
22	**circadian cycle**	实验证明，单独的SCN细胞仍然能够维持一个_____。	**第F段：** *Researchers at Massachusetts General Hospital isolated nerve cells from the SCN, and found they kept up a circadian cycle without help from daylight.* 马萨诸塞州总医院的研究人员从SCN中分离出神经细胞，发现这些细胞即使没有日光帮助，也仍然维持着一个昼夜节律周期。	题干中的： • **individual SCN cells**＝从SCN中被 **isolated** 出来的神经细胞； • **was maintained**＝*kept up*； • **experiments in the dark**＝*without help from daylight*。 空格前有冠词 **a**，需要填一个单数可数名词词组，答案为 **circadian cycle**。

Questions 23–24 多选题

题干翻译

下列哪些问题被文章认为与人体内部“生物钟”有关？选择两项。

正确答案：23 B，24 C

选项	判断	选项翻译	详细定位与翻译	详细解释
A	×	在没有窗户引入自然光的工作场所，产出下降	第F段只说： *The SCN senses daylight … which keep body functions in synchronization with the time of day.* SCN能够感知日光，并使身体功能与一天中的时间保持同步。	原文确实提到自然光会影响生物钟，但**没有提到没有窗户的工作场所**，也没有说这些工作场所的**产出下降**。 这是把“日光与生物钟有关”过度推断成“没有窗户导致生产率下降”。
B	✓	夜班工人无法正常完成任务	**第B段：** *When an individual's 'clock' is knocked out of synchronization through having to work at night or go on long-haul flights, that person can be left utterly unable to think or act.* 当一个人由于夜间工作或长途飞行而导致生物钟失去同步时，这个人可能会完全无法正常思考或行动。	**workers on night-shifts** 对应 **having to work at night**；**failure … to perform tasks** 对应 **utterly unable to think or act**。 虽然原文没有逐字写“完成任务失败”，但一个人无法思考或行动，自然就无法正常完成工作任务。
C	✓	一天中的特定时段，道路伤亡人数增加	**第B段：** *Traffic accident statistics show two deadly peaks, at 4 a.m. and again twelve hours later when people are at their least alert.* 交通事故统计显示，每天会出现两个致命事故高峰：凌晨4点，以及12小时后的另一个时段；这些时段人们的警觉性最低。	**at specific times of day** 对应凌晨4点以及12小时后；**an increase in casualties** 对应 **two deadly peaks**。 这里不是只说“事故数量增加”，而是使用了 **deadly**，明确涉及致命事故或道路伤亡。
D	×	在若干固定周期中的某些日子，人的效率下降	**第C段：** *They were supposed to result in 'critical days' when one or more cycles led to sub-optimal performance.* 据称，这些周期会产生所谓的“危险日”，在这些日子里，一个或多个周期会导致人的表现低于最佳水平。 但第D段紧接着说： *Scientists have since dismissed biorhythms as pseudoscience.* 科学家后来已经把这种生物节律理论斥为伪科学。	这是最有迷惑性的错误选项。它确实对应C段所谓23、28、33天固定周期及“表现下降”，但这些属于后来被否定的**biorhythms伪科学理论**，不是文章认可的24小时内部生物钟所造成的问题。 原文使用 **were supposed to**，表示“据称、被认为”，而不是作者承认的事实。
E	×	情绪波动随着太阳出现而加剧	原文只说SCN能够感知日光，并据此调节激素和身体功能；没有提到： • 太阳出现导致情绪波动； • 情绪波动上升。	选项利用了文章中的两个词汇元素：**Sun/daylight** 和C段中的 **emotional traits**，但两者并没有在原文中形成因果关系。属于拼接不同段落信息。

Questions 25–26 多选题

题干翻译

根据文章内容，下列哪些有关人体内部“生物钟”的问题仍然有待解决？选择两项。

正确答案：25 C，26 D

选项	判断	选项翻译	详细定位与翻译	详细解释
A	×	不同动物自由运行日周期的精确长度	**第E段：** *Organisms settle down to ‘free-running’ cycles that are close to, but rarely exactly, 24 hours long. In the case of humans, the cycle is typically around 24.5 hours long.* 生物会进入接近24小时、但很少恰好为24小时的自由运行周期；人类通常约为24.5小时。	原文说明不同生物的自由运行周期可能略有不同，但没有把“测定不同动物的精确周期长度”提出为一个**尚未解决的问题**。 多选题必须选择文章明确表示 **no-one knows / still being investigated** 的问题，不能因为文章未给出每一种动物的精确数字，就自行推断它仍有待解决。
B	×	生物钟暂时失去同步的原因	**第B段：** *When an individual’s ‘clock’ is knocked out of synchronization through having to work at night or go on long-haul flights…* 当一个人的生物钟由于夜间工作或长途飞行而失去同步时……	原文已经给出了造成失去同步的典型原因： · **work at night**； · **go on long-haul flights**。 因此，“为什么会暂时失去同步”并不是文章明确提出的未解问题。
C	✓	生物为什么会与太阳的节律建立联系	**第G段：** *Why did organisms bother to acquire a link with sunlight? So far, no-one knows, though there are several theories.* 生物为什么会特意形成与阳光之间的联系？到目前为止，没有人知道答案，尽管已经存在几种理论。	几乎是原文直接改写： · **the reason for**＝*why*； · **living beings**＝*organisms*； · **forming a connection**＝*acquire a link*； · **the Sun’s rhythms**＝*sunlight*所提供的昼夜变化。 后面的 **no-one knows** 明确表明这是未解决的问题。
D	✓	一个生物体内部的所有“生物钟”如何彼此保持同步	**第G段：** *But how do all the biochemical ‘clocks’ inside their cells stay in synchronization …? So far, no-one knows.* 但是，细胞内部所有的生化“时钟”究竟如何保持同步？到目前为止，没有人知道。 后文还说： *Sunlight might be useful in keeping the myriad cellular ‘clocks’ in lockstep.* 阳光可能有助于使数量众多的细胞生物钟保持步调一致。	题干的 **stay in step with one another** 对应： · **stay in synchronization**； · **in lockstep**。 文章虽然提出阳光“可能”帮助细胞生物钟同步，但使用的是 **might**，只是一个理论，并非确定答案，所以这一问题仍未解决。
E	×	在所有不同类型的生物中，“生物钟”具体位于何处	**第F段：** *There are ‘clocks’ inside each of their cells.* 每个细胞内部都存在“生物钟”。 **第G段：** *We know that circadian clocks exist throughout our bodies.* 我们已经知道昼夜生物钟遍布人体。 **第H段：** *Bacteria, fungi, plants and animals all appear to have evolved clock systems.* 细菌、真菌、植物和动物似乎都独立进化出了生物钟系统。	文章已经明确强调生物钟具有广泛的**细胞性分布**，并存在于多类生物中。更重要的是，文章没有把“所有生物中生物钟的具体位置”列为尚未解决的问题。 未解问题必须对应G段明确提出的两个问句，即C和D。